SANTIAGO RESTREPO RUIZ

Principal AI Engineer | Optimization, Cryptography & Electronics

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Oslo, Norway | From Colombia



PROFILE

Computer scientist and electronic engineer, specialized in numerical control and representation learning, robotics, circuits, cryptography and reliable superintelligent systems.

EXPERIENCE

Principal System Architect (Artificial Intelligence)

PricewaterhouseCoopers (PwC)

May 2025 – July 2026 — Montevideo, Uruguay

- Leads AI system architecture for enterprise environments, aligning solution design with business strategy, security, and delivery constraints.
- Designs scalable data and application architectures for intelligent automation, analytics, and machine-learning workflows in regulated contexts.
- Translates complex technical and operational requirements into production-ready platforms, integration patterns, and implementation roadmaps.

Senior C++ Optimization Engineer

Globant

January 2023 – May 2025 — Remote, Colombia

- Stabilized and optimized a large-scale C/C++ platform for construction logistics, improving reliability and performance through applied mathematics and disciplined engineering practices.
- Led development of a high-performance real-time street-mapping and routing engine designed for global-scale deployment.

Senior Data Engineer (Banking & Retail)

Fourier Analytics

March 2021 – January 2023 — Remote, Colombia

- Designed and implemented data pipelines in C++ for high-volume banking and retail workloads.
- Built time-series forecasting and analytical models for operational and commercial decision-making.
- Secured APIs across teams while supporting compliance and data-protection requirements.

Lead Quantitative Analyst (Finance Engineering Lead)

AUB Capital

2020 – 2021 — Medellín, Colombia

- Led analytics and engineering work around a USD 10 million private-equity portfolio.
- Developed quantitative models and automated systems for low-risk global market instruments from concept through implementation.
- Built low-latency interfaces, secure Linux infrastructure, and forecasting workflows for financial engineering operations.

RPA Developer (Financial Process Automation)

Stratesys

March 2018 – June 2020 — Remote, Spain

- Delivered financial process automation (Automation Anywhere, UiPath, and Blue Prism).
- Developed supporting tooling and frameworks in Python, JavaScript, and PowerShell.
- Applied NLP and document-processing to multilingual classification and electronic-invoice extraction.

Founding Partner

Waajacu

January 2012 – Present — Colombia

- Founded and led an open-source organization focused on computational diplomacy, cryptography, optimization, and high-performance software engineering.
- Researches cryptographic protocols for cooperation, markets, law, and robotics.
- Explores entropy maximization, stochastic modeling, nonlinear dynamical systems, and optimal control.

EDUCATION

Electronic Engineering

Universidad del Quindío

February 2013 – September 2018 — Armenia, Colombia

Thesis: Design and Modeling of a Delta Robot.

Mathematics and Algorithms

Massachusetts Institute of Technology (OpenCourseWare)

February 2013 – September 2018 — Online

Independent study in algorithms, mathematics, and computer science through MIT OpenCourseWare.

SELECTED HIGHLIGHTS

- Top SABER PRO Engineering result (2018).
- Founder of www.waajacu.com: advanced engineering and research.
- Founder of Save the Bees and Seeds, an open-source software foundation.
- Thesis: Delta robot design and modeling.

CORE EXPERTISE

- High-Performance C/C++
- Robotics & Autonomous Systems
- Embedded Systems & Electronics
- Optimization & Optimal Control
- Mathematical Modeling & Simulation
- AI / Machine Learning Systems
- LLM & Agentic AI Architecture
- Applied Cryptography & Computational Diplomacy
- Real-Time Systems & Algorithms
- Data Engineering & Distributed Systems

TECHNICAL SKILLS

CORE STACK

Linux / Unix	●●●●●
Windows	●●●○○
C / C++	●●●●●
Python / JavaScript	●●●●●
Rust / Go / PHP	●●●○○
FPGA / Verilog / VHDL	●●●●○
SQL / NoSQL	●●●●○
Git | Docker / Kubernetes / Nginx	●●●●○
GDB / Valgrind	●●●○○
ADK / LangChain / LangGraph	●●●●○
PyTorch / TensorFlow	●●●●●

APPLIED DOMAINS

Mathematical Modeling	●●●●●
Entropy / Information Theory	●●●●○
Control / Geometry	●●●●○
Differential Privacy	●●●●○
Probability / Reinforcement Learning	●●●●○
Embedded Systems / Networking	●●●●○
Computational Diplomacy	●●●●○

LANGUAGES

Spanish	●●●●●
English	●●●●●
Chinese (Simplified)	●●●○○